Your Name in Disguise

Objective

The goal of this activity was to create a secret codename by converting my name into hexadecimal format and my birth year into binary format. This activity helped me better understand numbering systems and how they are used in cybersecurity and computing.

Step 1: Convert My Name to Lowercase

Original Name: Fazel Rahman
Lowercase Name: fazel rahman

Step 2: Find the ASCII Decimal Values

Each letter in my name was converted into its ASCII decimal value.

Character	ASCII Decimal
f	102
a	97
z	122
e	101
l	108
(space)	32
r	114
a	97
h	104

m	109
a	97
n	110

Step 3: Convert the ASCII Values into Hexadecimal

f = 102

102 ÷ 16 = 6 remainder 6
Hexadecimal = 66

a = 97

97 ÷ 16 = 6 remainder 1
Hexadecimal = 61

z = 122

122 ÷ 16 = 7 remainder 10
10 in hexadecimal is represented by A
Hexadecimal = 7A

e = 101

101 ÷ 16 = 6 remainder 5
Hexadecimal = 65

l = 108

108 ÷ 16 = 6 remainder 12
12 in hexadecimal is represented by C
Hexadecimal = 6C

Space = 32

32 ÷ 16 = 2 remainder 0
Hexadecimal = 20

r = 114

114 ÷ 16 = 7 remainder 2
Hexadecimal = 72

a = 97

97 ÷ 16 = 6 remainder 1
Hexadecimal = 61

h = 104

104 ÷ 16 = 6 remainder 8
Hexadecimal = 68

m = 109

109 ÷ 16 = 6 remainder 13
13 in hexadecimal is represented by D
Hexadecimal = 6D

a = 97

97 ÷ 16 = 6 remainder 1
Hexadecimal = 61

n = 110

110 ÷ 16 = 6 remainder 14
14 in hexadecimal is represented by E
Hexadecimal = 6E

Final Hexadecimal Name

fazel rahman =

66 61 7A 65 6C 20 72 61 68 6D 61 6E

Step 4: Convert Birth Year into Binary

Birth Year: 2002

To convert the birth year into binary, I repeatedly divided the number by 2 and recorded the remainders.

Division	Quotient	Remainder
2002 ÷ 2	1001	0
1001 ÷ 2	500	1
500 ÷ 2	250	0
250 ÷ 2	125	0
125 ÷ 2	62	1
62 ÷ 2	31	0
31 ÷ 2	15	1
15 ÷ 2	7	1
7 ÷ 2	3	1
3 ÷ 2	1	1
1 ÷ 2	0	1

After reading the remainders from bottom to top, the binary number is:

11111010010

So:

$2002 = 11111010010_2$

Final Secret Codename

Hexadecimal Name

66 61 7A 65 6C 20 72 61 68 6D 61 6E

Binary Birth Year

11111010010

Flowchart

Start
↓
Write name in lowercase
↓
Find ASCII decimal values
↓
Convert decimal values into hexadecimal
↓
Combine hexadecimal values
↓
Write birth year
↓
Divide by 2 repeatedly
↓
Record remainders
↓
Read remainders from bottom to top
↓
Get binary number
↓
End

Challenges Encountered

One challenge I had while completing this assignment was converting decimal numbers into hexadecimal because hexadecimal uses both numbers and letters. I had to remember that values above 9 are represented by letters such as A, C, D, and E. Another challenge was keeping track of the remainders during the binary conversion process. At first, it was easy to lose track of the order, but after carefully checking my work and reading the remainders from bottom to top, I was able to get the correct binary value. Overall, this activity helped me better understand how different numbering systems are used in technology and cybersecurity.